Exercise 1

01 SELECT *

FROM employees;

Output:

| id | first_name | last_name | department | Salary | hire_date | City |
|---|---|---|---|---|---|---|
| 1 | Alice | Green | IT | 70000 | 2020-01-10 | Johannesburg |
| 2 | Brian | lee | HR | 45000 | 2019-03-22 | Cape Town |
| 3 | Cathy | Zulu | Finance | 65000 | 2018-04-18 | Durban |
| 4 | David | Mokoena | Marketing | 50000 | 2020-11-05 | Pretoria |
| 5 | Eva | Naidoo | IT | 72000 | 2017-09-30 | Johannesburg |

02 SELECT DISTINCT department

FROM employees;

out: Department

IT

HR

Finance

Marketing

03 SELECT first_name, last_name, salary

FROM employees

ORDER BY Salary DESC;

output:

| first_name | last_name | salary |
|---|---|---|
| Eva | Naidoo | 72000 |
| Alice | Green | 70000 |
| Cathy | Zulu | 65000 |
| David | Mokoena | 50000 |
| Brian | Lee | 45000 |

04 SELECT id, first_name, last_name, salary
FROM employees
ORDER BY salary DESC LIMIT 3;

Output:

| id | first_name | last_name | salary |
|---|---|---|---|
| 5 | Eva | Naidoo | 720000 |
| 1 | Alice | Green | 70000 |
| 3 | Cathy | Zulu | 65000 |

05 SELECT id, first_name, last_name, department
FROM employees
WHERE department = 'IT';

Output:

| id | first_name | last_name | department |
|---|---|---|---|
| 1 | Alice | Green | IT |
| 5 | Eva | Naidoo | IT |

06 SELECT id, first_name, last_name, department, salary
FROM employees
WHERE department = 'Finance' AND salary > 6000;

Output:

| id | first_name | last_name | department | salary |
|---|---|---|---|---|
| 3 | Cathy | Zulu | finance | 65000 |

07 SELECT id, first_name, last_name, department
FROM employees
WHERE department = HR OR department = Marketing;

Output:

| id | first_name | last_name | department |
|---|---|---|---|
| 2 | Brian | lee | HR |
| 4 | David | Moroena | Marketing |

08 SELECT id, first_name, last_name, department
FROM employees
WHERE department != 'IT';

Output:

| id | first_name | last_name | department |
|---|---|---|---|
| 2 | Brian | lee | HR |
| 3 | Cathy | Zulu | finance |
| 4 | David | Moroena | Marketing |

09 SELECT id, first_name, last_name, department
FROM employees
WHERE department IN('IT', 'HR', 'Finance');

Output:

| id | first_name | last_name | department |
|---|---|---|---|
| 1 | Alice | Green | IT |
| 2 | Brian | lee | HR |
| 3 | Cathy | Zulu | Marketing |
| 5 | Eva | Naidoo | IT |

10 SELECT id, first_name, last_name, department, salary, city
FROM employees
WHERE department='IT'
AND salary>65000
AND city = 'Johannesburg';

Output:

| id | first_name | last_name | department | salary | city |
|---|---|---|---|---|---|
| 1 | Alice | Green | IT | 70000 | Johannesburg |
| 5 | Eva | Naidoo | IT | 72000 | Johannesburg |